

| Learning outcomes                                                                                                                            | Additional guidance                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p><i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i></p>                                         |                                                                                      |
| (a) potential divider circuit with components                                                                                                | Learners will also be expected to know about a potentiometer as a potential divider. |
| (b) potential divider circuits with variable components e.g. LDR and thermistor                                                              |                                                                                      |
| (c) (i) potential divider equations e.g.                                                                                                     | M2.3                                                                                 |
| $V_{\text{out}} = \frac{R_2}{R_1 + R_2} \times V_{\text{in}} \quad \text{and} \quad \frac{V_1}{V_2} = \frac{R_1}{R_2}$                       |                                                                                      |
| (c) (ii) techniques and procedures used to investigate potential divider circuits which may include a sensor such as a thermistor or an LDR. | <p><b>PAG4</b></p> <p>HSW6 Designing temperature and light sensing circuits.</p>     |

